MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. How many two-digit numbers are divisible by 3? [1]

(A) 25 (B) 30 (C) 32 (D) 33

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 30** [0.5 Mark]

Given: Two-digit numbers divisible by 3 form the AP $12, 15, 18, \ldots, 99$, with $a = 12$, $d = 3$, and $a_n = 99$.

We use the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$$12 + (n-1)(3) = 99 \implies 3(n-1) = 99 - 12 = 87 \implies n - 1 = \frac{87}{3} = 29 \implies n = \mathbf{30}$$

Q2. If the nth term of an AP is $a_n = 2n + 1$, then the sum of its first n terms S_n is: [1]

(A) $n(n-2)$ (B) $n(n+2)$ (C) $n(n+1)$ (D) n^2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $\mathbf{n(n+2)}$** [0.5 Mark]

Given: $a_n = 2n + 1$, so first term $a = a_1 = 2(1) + 1 = 3$ and last term $l = a_n = 2n + 1$.

We use the first-and-last-term sum formula:

Formula / Theorem Used: $S_n = \frac{n}{2}(a + l)$

Justification: [0.5 Mark]

$$S_n = \frac{n}{2}(3 + (2n+1)) = \frac{n}{2}(2n+4) = \frac{n \times 2(n+2)}{2} = \mathbf{n(n+2)}$$

Q3. The common difference of the AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots$ $(p \neq 0)$ is: [1]

(A) p (B) $-p$ (C) -1 (D) 1

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(C) -1** [0.5 Mark]

We apply the common difference definition:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$$d = \frac{1-p}{p} - \frac{1}{p} = \frac{(1-p)-1}{p} = \frac{-p}{p} = \mathbf{-1}$$

Q4. If the nth term of an AP is $a_n = 3 - 4n$, then its common difference is: [1]

(A) 3 (B) 4 (C) -4 (D) -3

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(C) -4** [0.5 Mark]

We evaluate consecutive terms a_1 and a_2:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = 3 - 4(1) = -1$ and $a_2 = 3 - 4(2) = -5$. Thus $d = -5 - (-1) = -5 + 1 = \mathbf{-4}$.

Q5. **Assertion (A):** The 10th term from the end of the AP $4, 9, 14, \ldots, 254$ is 209.
Reason (R): In an AP with last term l and common difference d, the nth term from the end is given by $l - (n-1)d$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(B) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

We apply the backward AP formula:

Formula / Theorem Used: n^{th} term from end $= l - (n-1)d$

Justification: [0.5 Mark]

Here $l = 254$, $d = 9-4 = 5$, and $n = 10$. Thus 10th term from end $= 254-(10-1)(5) = 254-45 = 209$. Hence (A) and (R) are both true and (R) directly proves (A).

Q6. In an AP, the first term is 5 and the sum of its first four terms is half the sum of the next four terms. Find the common difference. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: First term $a = 5$. Sum of first 4 terms is S_4; sum of next 4 terms $(a_5 + a_6 + a_7 + a_8)$ is $S_8 - S_4$.
We use the AP sum formula:

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 1: Translating the Condition into an Equation [1 Mark]
Given $S_4 = \frac{1}{2}(S_8 - S_4) \implies 2S_4 = S_8 - S_4 \implies 3S_4 = S_8$.

Step 2: Substituting and Solving for d [1 Mark]

$$3 \cdot \frac{4}{2}[2(5) + 3d] = \frac{8}{2}[2(5) + 7d] \implies 6(10 + 3d) = 4(10 + 7d)$$

$$60 + 18d = 40 + 28d \implies 60 - 40 = 28d - 18d \implies 20 = 10d \implies d = \boxed{2}$$

Q7. Find the sum of the first 15 terms of an AP whose n^{th} term is given by $a_n = 9 - 5n$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Finding the First Term and 15th Term [1.5 Marks]
Given: General term $a_n = 9 - 5n$.
Putting $n = 1$: $a_1 = 9 - 5(1) = 9 - 5 = 4$.
Putting $n = 15$: $a_{15} = 9 - 5(15) = 9 - 75 = -66$.
(Also check $a_2 = 9 - 10 = -1 \implies d = -1 - 4 = -5$, constant.)

Formula / Theorem Used: $S_n = \frac{n}{2}(a_1 + a_n)$

Step 2: Computing S_{15} [1.5 Marks]

$$S_{15} = \frac{15}{2}(a_1 + a_{15}) = \frac{15}{2}(4 - 66) = \frac{15}{2} \times (-62) = 15 \times (-31) = \boxed{-465}$$

Q8. If the sum of the first n terms of an AP is $S_n = 2n^2 + 3n$, find its 4th term. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $S_n = 2n^2 + 3n$. We use the difference identity between consecutive partial sums:

Formula / Theorem Used: $a_n = S_n - S_{n-1} \qquad (n \geq 2)$

Step 1: Evaluating S_4 and S_3 [1.5 Marks]
Putting $n = 4$: $S_4 = 2(4)^2 + 3(4) = 2(16) + 12 = 32 + 12 = 44$.
Putting $n = 3$: $S_3 = 2(3)^2 + 3(3) = 2(9) + 9 = 18 + 9 = 27$.

Step 2: Finding a_4 [1.5 Marks]

$$a_4 = S_4 - S_3 = 44 - 27 = \boxed{17}$$

Q9. Find the sum of the first 20 terms of an AP in which $d = 7$ and the 20th term is 143. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Number of terms $n = 20$, common difference $d = 7$, and $a_{20} = 143$.
We use the general term and sum formulas:

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

Step 1: Finding the First Term a [1.5 Marks]

$$a_{20} = a + (20-1)(7) = 143 \implies a + 19 \times 7 = 143 \implies a + 133 = 143 \implies a = 143 - 133 = 10$$

Step 2: Calculating S_{20} [1.5 Marks]

$$S_{20} = \frac{20}{2}(a + a_{20}) = 10(10 + 143) = 10 \times 153 = \boxed{\mathbf{1{,}530}}$$

MATHS MANTHAN TUITION CLASSES OFFICIAL SOLUTIONS

Q10. CASE STUDY: Annual Sports Day Lemon-and-Bucket Race

In a lemon-and-bucket race held during a school's annual sports day, a bucket is placed at the starting point, which is 5 m from the first lemon L_1, and the remaining lemons $L_2, L_3, \ldots, L_{10}$ are placed 3 m apart in a straight line. There are 10 lemons in total. A competitor starts from the bucket, picks up the nearest lemon, runs back to drop it in the bucket, runs back to pick up the next lemon, and continues in this way until all 10 lemons are in the bucket: [4]

(i) What is the total distance run by the competitor to collect the 1st lemon? [1]

(ii) What is the total distance run by the competitor to collect the 2nd lemon? [1]

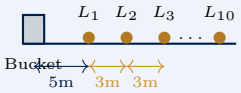
(iii) Write the AP of round-trip distances and find the total cumulative distance run by the competitor. [2]

— OR —

What distance does the competitor run exclusively to collect the 10th (final) lemon? [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: One-way distances are 5 m, 8 m, 11 m, $\ldots$. Each pickup requires a round trip ($2 \times$ one-way distance).



We apply the AP term and sum formulas:

Formula / Theorem
Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

Sub-part (i): Round-Trip Distance for 1st Lemon [1 Mark]

Running 5 m to L_1 and 5 m back to the bucket: $a_1 = 2 \times 5 = \boxed{\mathbf{10}\ \text{m}}$.

Sub-part (ii): Round-Trip Distance for 2nd Lemon [1 Mark]

Distance from bucket to L_2 is $5 + 3 = 8$ m. Round trip: $a_2 = 2 \times 8 = \boxed{\mathbf{16}\ \text{m}}$.

Sub-part (iii): AP and Total Distance for All 10 Lemons [2 Marks]

For L_3, round trip $= 2(5 + 3 + 3) = 22$ m. Thus the AP is **10**, **16**, **22**, $\ldots$ with $a = 10$, $d = 6$, and $n = 10$.

$$S_{10} = \frac{10}{2}[2(10) + (10-1)(6)] = 5(20 + 54) = 5 \times 74 = \boxed{\mathbf{370}\ \text{m}}$$

Sub-part (iii) [OR Option]: Distance Only for the 10th (Final) Lemon [2 Marks]

One-way distance to $L_{10} = 5 + 9(3) = 32$ m, so round-trip distance:

$$a_{10} = 2 \times 32 = 10 + (10-1)(6) = 10 + 54 = \boxed{\mathbf{64}\ \text{m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★